

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 1- INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA1223	Course Title:	Computer Architecture
CO Assessed:	CO4	Assessment:	ASSESSMENT TOOL1- INDUSTRY PROBLEM SOLVING TASK
Weightage:	30%	Total Marks:	25
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
Student Name:	Monal Raj. B	Reg.No.:	192511176
Department:	B.E - CSE	Date:	

ANNEXURE A
INDUSTRY PROBLEM SOLVING TASK

62% 48

- Smartphone Performance Optimization**
A mobile manufacturer observes lag while switching between apps. Analyze the role of L1, L2, L3 cache and DRAM and propose a hierarchical memory redesign to reduce latency and improve throughput.
- Cloud Server Memory Bottleneck**
A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.
- Autonomous Vehicle Data Processing**
An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.
- AI Inference Edge Device**
An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.
- High-Performance Gaming Console**
A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

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ANNEXURE A
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1. Smartphone Performance Optimization
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2. Cloud Server Memory Bottleneck
A cloud service provider experiences high latency in database queries. Compare cache hierarchy vs virtual memory paging and recommend improvements using appropriate memory technologies.
3. Autonomous Vehicle Data Processing
An autonomous vehicle must process sensor data in real time. Evaluate SRAM, DRAM, and MRAM and design a memory hierarchy that maximizes bandwidth and minimizes latency.
4. AI Inference Edge Device
An edge AI device needs fast model loading with low power consumption. Compare NAND Flash, DRAM, and RRAM and recommend an optimized hierarchical memory structure.
5. High-Performance Gaming Console
A gaming console suffers frame drops during large scene loading. Analyze L3 cache vs DRAM throughput and propose memory hierarchy enhancements.

1. Smartphone Performance Optimization

Aim:

To analyze the role of L1, L2, L3 cache and DRAM in smartphone performance and propose an optimized memory hierarchy to reduce latency and improve system throughput.

Explanation:

Modern smartphones are designed to handle multiple applications simultaneously, such as social media, gaming, and background processes. The performance of these devices mainly depends on how efficiently the processor accesses memory. Memory in a smartphone is organized in a hierarchical manner consisting of L1 cache, L2 cache, L3 cache, and DRAM.

The L1 cache is the smallest and fastest memory located inside the processor core. It stores frequently used instructions and data, allowing extremely fast access. The L2 cache is larger than L1 but slightly slower, and it acts as an intermediate storage between L1 and L3. The L3 cache is shared among multiple cores and has a larger capacity but relatively higher latency. Finally, DRAM is the main memory which has a very large storage capacity but significantly higher access time compared to cache memory.

When a user switches between applications, the processor needs to fetch data repeatedly. If the required data is available in cache (cache hit), the execution is fast. However, if the data is not present (cache miss), the processor must access DRAM, which introduces delay and causes lag.

Analysis:

The major issue in smartphone performance arises due to inefficient cache utilization and frequent memory access from DRAM. When applications are switched frequently, previously used data may be removed from cache, leading to increased cache misses. This results in higher latency and reduced system responsiveness. Moreover, improper scheduling of memory access further increases delays.

Proposed Solution:

To improve performance, the following enhancements can be implemented:

- Increase the size of L2 and L3 cache to store more application data
- Implement advanced cache replacement policies such as Least Recently Used (LRU)
- Use prefetching techniques to load data before it is requested
- Optimize DRAM scheduling to reduce access delay
- Implement Unified Memory Architecture (UMA) to allow better sharing of memory resources
- Use faster LPDDR memory for improved bandwidth

Conclusion/Result:

Thus, by improving cache efficiency and optimizing memory hierarchy, latency can be reduced significantly. This leads to smoother app switching, better multitasking performance, and overall improved smartphone responsiveness.

2. Cloud Server Memory Bottleneck

Aim:

To analyze the causes of memory bottlenecks in cloud servers and compare cache hierarchy with virtual memory paging, and propose suitable solutions to improve performance.

Explanation:

Cloud servers are designed to handle a large number of user requests simultaneously. These servers process complex database queries and store massive amounts of data. Memory performance plays a critical role in ensuring fast response times.

The system uses cache memory to store frequently accessed data so that it can be retrieved quickly. However, cache size is limited. When the required data is not found in cache, the system accesses main memory (RAM), and if RAM is full, it uses virtual memory paging, where data is stored on disk.

Paging significantly increases access time because disk storage (even SSDs) is much slower than RAM. This leads to high latency in query execution and reduces overall system performance.

Analysis:

The main bottleneck occurs due to excessive paging and inefficient use of cache memory. When multiple users access the system simultaneously, memory demand increases, leading to frequent swapping between RAM and disk. This increases response time and reduces throughput.

Cache memory improves speed but is limited in size, whereas virtual memory provides larger capacity but introduces delay. Therefore, balancing these two is critical.

Proposed Solution:

To overcome memory bottlenecks, the following techniques can be applied:

- Increase cache size and improve cache hit ratio
- Use high-speed SSDs instead of HDDs for paging
- Implement distributed caching systems such as Redis or Memcached
- Optimize database queries to reduce memory load
- Use in-memory databases for faster data access

- Apply intelligent memory allocation and load balancing

Conclusion/Result:

Thus, by reducing dependency on paging and improving cache efficiency, cloud server performance can be significantly enhanced. This results in lower latency, faster query execution, and improved user experience.

3. Autonomous Vehicle Data Processing

Aim:

To evaluate different memory technologies such as SRAM, DRAM, and MRAM, and design an efficient memory hierarchy for real-time autonomous vehicle data processing.

Explanation:

Autonomous vehicles rely heavily on real-time data processing to make decisions such as obstacle detection, lane tracking, and navigation. These vehicles use sensors like cameras, LiDAR, and radar, which generate a massive amount of data continuously. To process this data efficiently, a high-performance memory system is required.

Different types of memory technologies are used in such systems. SRAM (Static Random Access Memory) is very fast and is mainly used in cache memory inside the processor. It provides low latency but is expensive and has limited storage capacity. DRAM (Dynamic Random Access Memory) is slower than SRAM but offers higher storage capacity and is used as the main memory. MRAM (Magnetoresistive RAM) is an emerging memory technology that combines the speed of SRAM with the non-volatility of storage memory, meaning it can retain data even when power is off.

In autonomous systems, data must be processed instantly without delay. Therefore, the memory system must support high bandwidth and low latency.

Analysis:

The main challenge is balancing speed, storage capacity, and power consumption. SRAM is very fast but cannot store large amounts of data. DRAM can store large data but introduces latency. MRAM provides a balance but is still developing and costly. A single type of memory cannot meet all requirements efficiently.

Proposed Solution:

An optimized hybrid memory hierarchy can be designed as follows:

- Use SRAM as cache memory for storing frequently accessed data
- Use DRAM as main memory for storing large volumes of sensor data
- Use MRAM for persistent storage to retain critical data even during power failure

- Implement parallel processing techniques to increase data throughput
- Optimize memory bandwidth to support real-time processing

Conclusion/Result:

Thus, by combining SRAM, DRAM, and MRAM in a hierarchical structure, autonomous vehicles can achieve fast, reliable, and efficient data processing, ensuring safe and accurate real-time decision-making.

4. AI Inference Edge Device

Aim:

To design an optimized memory hierarchy for an AI inference edge device that ensures fast model execution with low power consumption.

Explanation:

AI edge devices are systems that perform machine learning tasks locally without relying on cloud servers. Examples include smart cameras, voice assistants, and IoT devices. These devices must process data quickly while consuming minimal power.

Different memory technologies are used in such systems. NAND Flash is used for storing large AI models due to its high storage capacity, but it has slower access speed. DRAM is used for faster data access during execution but consumes more power. RRAM (Resistive RAM) is a new memory technology that offers fast access speed and low power consumption.

When an AI model is executed, it must be loaded from storage into faster memory for processing. This process can introduce delay if not optimized.

Analysis:

The main challenges in AI edge devices are reducing power consumption while maintaining high performance. Large AI models increase memory usage, and frequent memory access increases latency. Using only DRAM leads to higher energy consumption, while relying only on NAND Flash results in slower performance.

Proposed Solution:

An efficient memory hierarchy can be designed as follows:

- Store AI models in NAND Flash for high-capacity storage
- Load active data into DRAM for fast processing
- Use RRAM for intermediate computations due to its speed and low power usage
- Apply model compression techniques to reduce memory requirements

- Use intelligent data management to reduce unnecessary memory access

Conclusion/Result:

Thus, by combining NAND Flash, DRAM, and RRAM, AI edge devices can achieve faster inference speed while maintaining low power consumption, making them efficient and reliable for real-time applications.

5. High-Performance Gaming Console

Aim:

To analyze the impact of L3 cache and DRAM on gaming performance and propose improvements in memory hierarchy to reduce frame drops.

Explanation:

High-performance gaming consoles require efficient memory systems to handle complex graphics and large game environments. During gameplay, a large amount of data such as textures, models, and physics calculations must be processed continuously.

The L3 cache is a high-speed memory that stores frequently accessed data and is shared among processor cores. It provides faster access compared to main memory but has limited storage capacity. DRAM, on the other hand, provides large storage capacity but has higher latency compared to cache memory.

When the processor cannot find required data in the cache (cache miss), it must fetch data from DRAM. This delay can cause frame drops and lag during gameplay, especially in large open-world games.

Analysis:

The main issue is the imbalance between speed and capacity. L3 cache is fast but small, while DRAM is large but slower. During high-demand scenarios such as scene loading, large data must be fetched from DRAM, increasing latency. This results in reduced frame rates and poor gaming experience.

Proposed Solution:

To improve performance, the following enhancements can be implemented:

- Increase the size of L3 cache to store more frequently used data
- Use high-speed memory such as GDDR or HBM DRAM
- Implement cache prefetching techniques to load data in advance
- Optimize memory bandwidth to support high data transfer rates
- Use parallel memory access techniques for faster processing

Conclusion/Result:

Thus, by improving cache efficiency and using high-speed DRAM, gaming consoles can reduce latency and eliminate frame drops, resulting in smoother gameplay and enhanced user experience.

Code of Conduct:

I Monal Raj.B (192511176) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

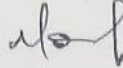
	Understanding of Industry Problem (20%)	Application of Engineering Concepts (25%)	Solution Design (25%)	Use of Tools (15%)	Analysis (10%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints(1)	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem(1.25)	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards(1.25)	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data(0.75)	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation(0.5)	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

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	Understanding of Industry Problem (20%)	Application of Engineering Concepts (25%)	Solution Design (25%)	Use of Tools (15%)	Analysis (10%)
Q1	.5	1	1	.5	.5
Q2	.5	1	1	.5	.5
Q3	1	.5	1	.5	.5
Q4	0.5	.5	.5	.5	.5
Q5	.5	1	.5	.5	.5

3.5
3.5
3.5
2.5
2.5

15.5

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